PM12: four-channel CFD and TDC characterization

Incremental test chapter — 10 November 2026

Board: `PM12-K7-3A17A1B402212223`; EEPROM serial 1803.
Populated channels: 12, 10, 8 and 6.
Run: `pm12-cfd-four-20261110`.

Measured outcome

The four TDC input paths passed finite-event and phase-response testing. Every one of 25 input phases produced exactly 128 records on the selected channel, with no events on the other eleven channels. All eight quiet windows were empty. The scan contains 12,800 events, plus 64 mapping-discovery events.

PM channel	RJ45 index	Slope [fine codes/ns]	Max. spread [fine codes RMS]
12	0	-46.185	1.147
10	1	-46.201	1.185
8	2	-46.194	1.029
6	3	-46.177	1.083

Slope magnitudes differ by approximately 0.051%. Timing is relative to the programmed Kasli delay, assuming a 40.0787 MHz source. This is not an independent calibration of absolute TDC timestamps or an intrinsic jitter measurement.

Fixture connections

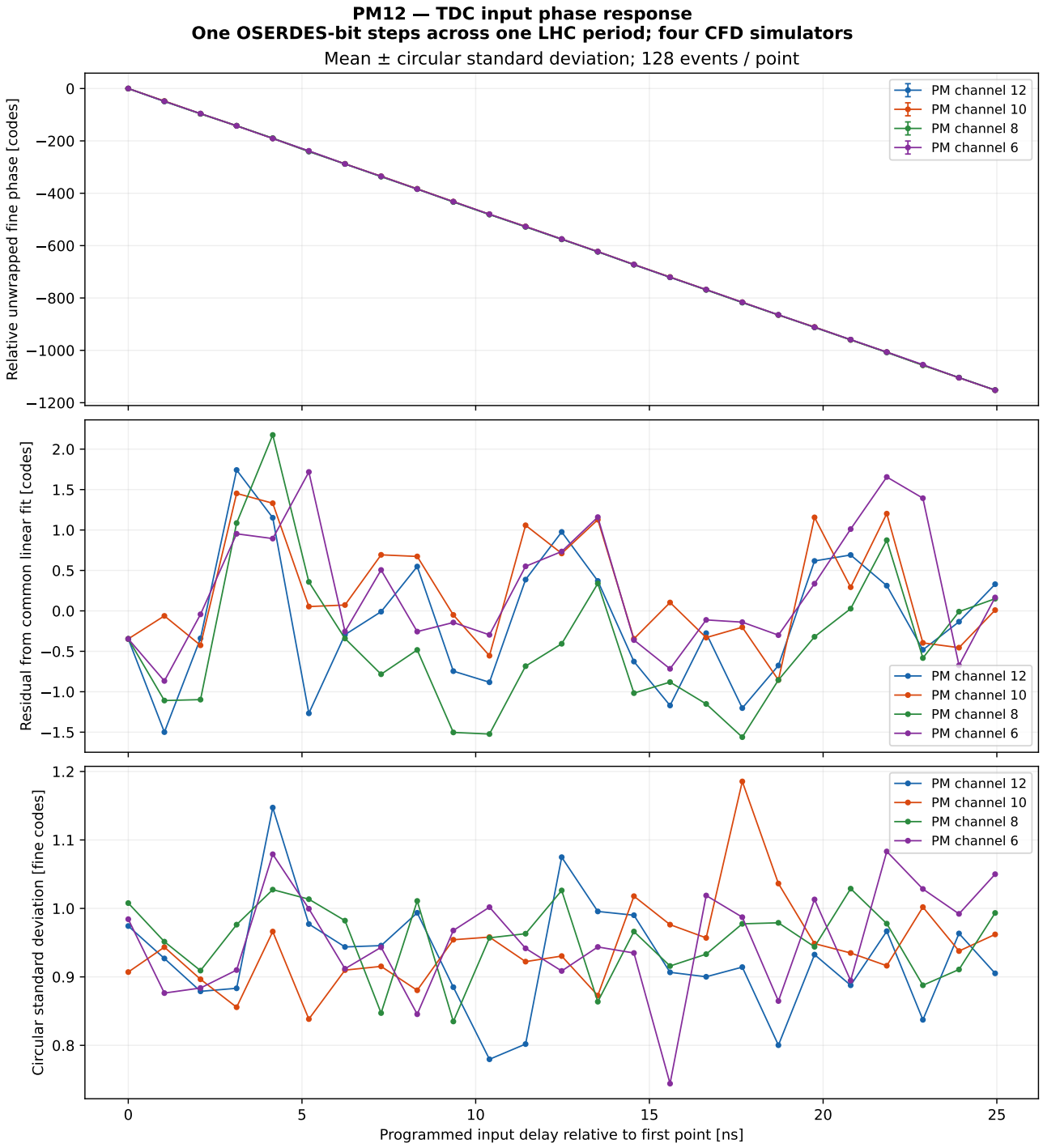
PM channel	DIOT slot	RJ45 index	CFD TX channel	Gate RX A/B
12	7	0	0	8/9
10	7	1	4	10/11
8	7	2	8	12/13
6	7	3	12	14/15

Indices are zero based; Kasli occupies slot 0. HDMI remains on slot 6. PM12 and Kasli use the shared LHC-derived reference. SFP0 is Kasli control LAN, SFP1 the private PM link, and SFP2 the PM LAN bridge.

Acquisition procedure

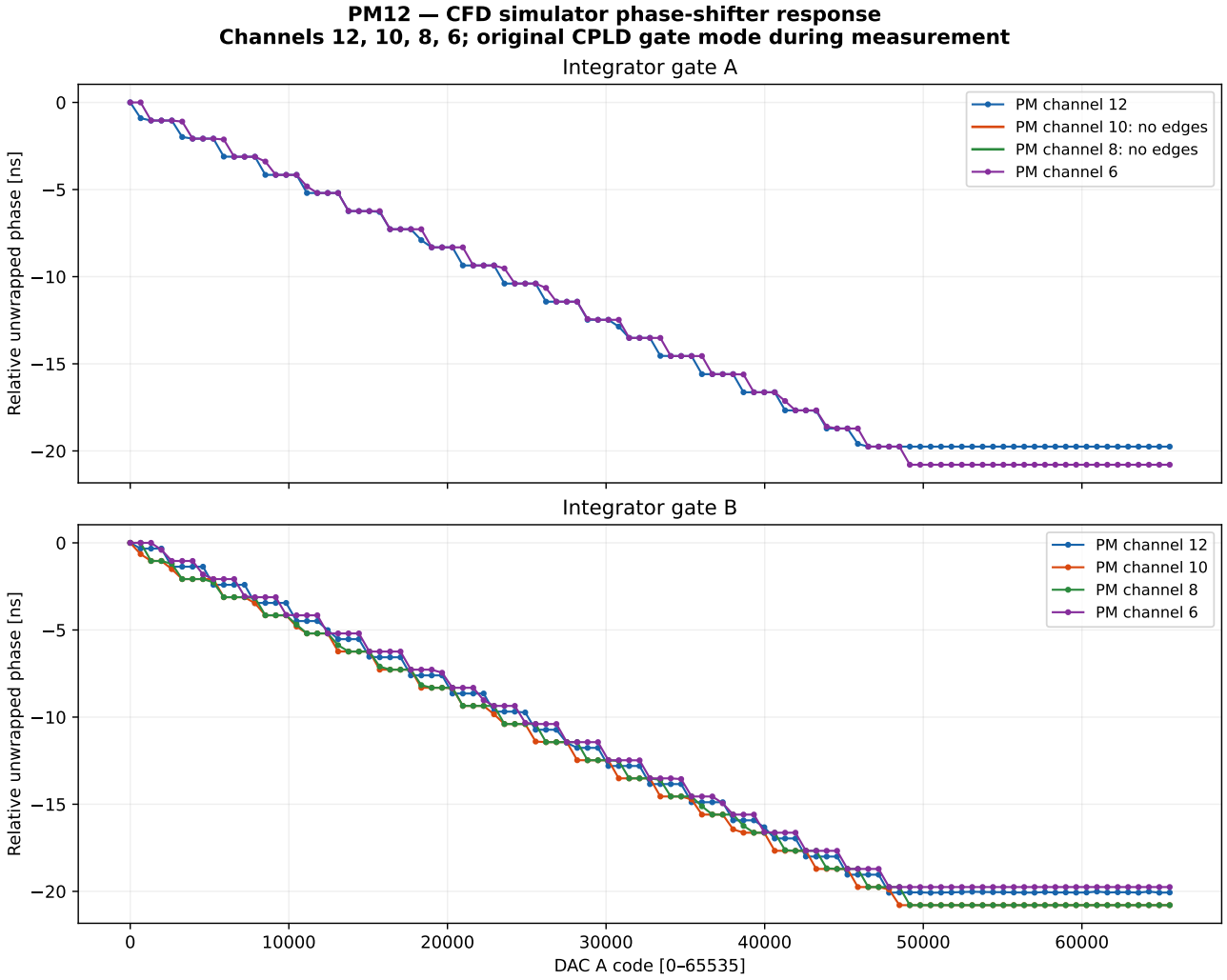
Check board identity and physical PGOOD; inhibit shared analog monitors. Discover each CFD input with 16 pulses. Scan 25 serializer phases with 128 pulses each, one bit (approximately 1.0396 ns) per step, across 24.951 ns. Freeze TDC buffers and download all channels through IPbus. Separately, sweep each internal DAC-A through 101 codes from 0 to 65535, recording all eight gate-return lanes with the Kasli ISERDES recorder. The external MOD-DAC-28 and shared DMM were not used.

TDC input phase response



Raw fine phase is unwrapped modulo 128 by nearest continuation. The middle panel shows residuals from one common linear fit, exposing differences hidden by the overlapping response curves. Coarse fields are retained separately.

Internal DAC-A phase-shifter response



Observed range: approximately 20–21 ns, with a plateau near the upper part of the DAC range (around code 48,000). The sampler bins are approximately 1.04 ns; fractional centroids do not establish calibrated sub-bin accuracy.

Unresolved return-path findings: INTA for PM channels 10 and 8 stayed HIGH throughout the sweep. Their INTB returns respond to the corresponding DAC. These coincide with slot7 DIO channels 5 and 9, already unqualified in the 16 October loopback (5↔9). The 28 October dynamic loopback also failed 9↔13. This is evidence to check the LVDS fixture before assigning a PM/CFD fault. The CPLD source generates INTA on both channels; only INTB12 was intentionally suppressed. Historical correlation added after the initial report; no new endpoint-isolation measurement was performed.

On channel 12, the prior INTB-off experiment suppressed the measured DAC-dependent gate movement. Original complementary gates were temporarily restored for this scan; the INTB-off CPLD12 image was restored and vendor-verified afterwards.

Evidence and retained hardware state

Data integrity

The 404 full-range DAC points contain 105,880,320 raw sampled bits. During acquisition, 4,411,680 hardware phase records were checked against the raw samples. All TDC raw counts and fine histograms were independently checked against the downloaded binary files. The legacy single-port test also passed with exactly 16 events on channel 12 after the four-port scan.

Build qualification

The Kasli extension uses four independent output serializers and preserves 18 input serializers, HDMI and the three active SFP transceiver channels. Dell Vivado 2025.2 qualification: setup slack 0.136 ns, hold slack 0.068 ns, zero DRC errors. Physical CFD pin pairs were checked explicitly.

PM diagnostic bitstream SHA-256:
`1dbafd99033f803a086d4153ed3d873adfc9b4208b806785f00d3024bdfbf1ac`

Kasli four-port bitstream SHA-256:
`a71a08a258620b0e5a52c5d03aefd8c69040c9d15e8de6391c484ac182b7173b`

Final retained configuration

PM image c7c42d28 and the new four-port Kasli image remain in FPGA SRAM. CPLD12 retains INTB-off; the other eleven CPLD images are unchanged. Internal DAC-A codes for channels 12/10/8/6 were explicitly restored to zero. All shared monitor multiplexers are inhibited and both upstream I2C switches are disconnected. CFD/HDMI outputs are off, port selection is zero, and both DIO cards are inputs. TDC buffers are frozen and empty. ADC was not acquired after restoration of the PM SRAM image. PGOOD and fresh FPGA DNA passed.

No MCU firmware, EEPROM, FPGA flash or PLL profile was written. The intended CPLD12 gate-mode changes used vendor erase/program/verify procedures. No external analog source or DMM was used. All hardware leases were released.

Editable sources and raw data

This `report.tex` and both vector PDF/SVG figures are editable. `gate-phase.csv`, `tdc-phase.csv` and `summary.json` contain plot inputs and statistics. `raw-and-procedures.tar.gz` retains binary buffers, raw phase captures, commands and exact executed procedures. `gateware-artifacts.tar.gz` retains generated HDL, source snapshots, qualified bitstreams, recovery images and qualification reports. File hashes are listed in `sha256.json`.